Session 004 Initial Data: A Two-Clump Perturbation of an $M_{\rm env} = 0.76$ Einstein–Vlasov Equilibrium

GI-in-cluster campaign

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Abstract

We independently construct the Session-004 initial slice: a finite-radius spherical Einstein–Vlasov equilibrium with ADM mass 0.76 and areal radius 30, plus two cold Gaussian conformal sources of mass 0.12 at $x = \pm 3$. Conformal flatness and $K_{ij} = 0$ make the Hamiltonian solve linear and give continuum ADM mass one. We give the component-dependent five-million-particle realization, the recalculated Jeans and horizon scales, the compact AMR hierarchy, and the actual Z4c/gauge choices. A separate construction gives opposite orthonormal y boosts for orbital angular momentum. It preserves the Hamiltonian normalization but introduces a local momentum current which $K_{ij} = 0$ does not balance; its measured constraint gate is reported explicitly.

1 Static spherical envelope

In areal radius r use

$$\mathrm{d}s^2 = -e^{2\mu(r)}\mathrm{d}t^2 + \left(1 - \frac{2m(r)}{r}\right)^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2. \tag{1}$$

For orthonormal spatial four-velocity $\boldsymbol{v}$, $W = \sqrt{1+|\boldsymbol{v}|^2}$ and particle energy $E = e^{\mu}W$. The isotropic cutoff ansatz is newly solved, not rescaled from the old $M_0 = 0.61$ table:

$$f_0(E) = A(E_{\rm cut} - E)_+, \quad \rho_0 = 4\pi\int_0^{v_{\max}} f_0 W v^2 \mathrm{d}v, \quad p_0 = \frac{4\pi}{3}\int_0^{v_{\max}} f_0 \frac{v^4}{W}\mathrm{d}v. \tag{2}$$

Writing $y = \log E_{\rm cut} - \mu$ and $B = AE_{\rm cut}$ gives $v_{\max} = \sqrt{e^{2y}-1}$. The shooting equations are

$$m' = 4\pi r^2\rho_0(y), \qquad y' = -\frac{m + 4\pi r^3 p_0(y)}{r(r-2m)}, \qquad m(0) = 0, \quad y(0) = y_c, \tag{3}$$

and the first zero $y(R_0) = 0$ is matched to Schwarzschild,

$$E_{\rm cut} = e^{\mu(R_0)} = \sqrt{1 - 2M_{\rm env}/R_0}. \tag{4}$$

For $(M_{\rm env}, R_0) = (0.76, 30)$ the converged result is

y_c	0.0705113472056
$E_{\rm cut}$	0.974337381677
A, B	0.0264154946062, 0.0257376038503
rest mass $M_{\rm rest}$	0.771406673590
central ρ_0, p_0	1.7583951×10^{-4}, 3.5275987×10^{-6}

The code's independent conformal-source integral is $\int 4\pi R^2 q_0 \mathrm{d}R = 0.7599999985$. Radial-step convergence is

Δr	y_c
0.004	0.070511361859
0.002	0.070511351402
0.001	0.070511347206
0.0005	0.070511346737

1.1 Isotropic radius and conformal fields

The isotropic radius R obeys

$$\frac{\mathrm{d}R}{R} = \frac{\mathrm{d}r}{r\sqrt{1-2m/r}}, \qquad r = \psi_0^2 R, \qquad R(R_0) = \frac{R_0 - M_{\rm env} + \sqrt{R_0(R_0-2M_{\rm env})}}{2}. \tag{5}$$

Numerically $R(R_0) = 29.2350607252$, $\psi_0(0) = 1.04808910510$, and $\alpha_0(0) = 0.908001730301$. Define

$$q_0(R) = \psi_0^5 \rho_0, \qquad \gamma_{ij}^{(0)} = \psi_0^4 \delta_{ij}, \qquad K_{ij}^{(0)} = 0, \tag{6}$$

so $\nabla^2 \psi_0 = -2\pi q_0$. Figure 1 records the solved profiles, areal–isotropic map, enclosed mass, fixed-microscopic-ansatz sequence, and new source.

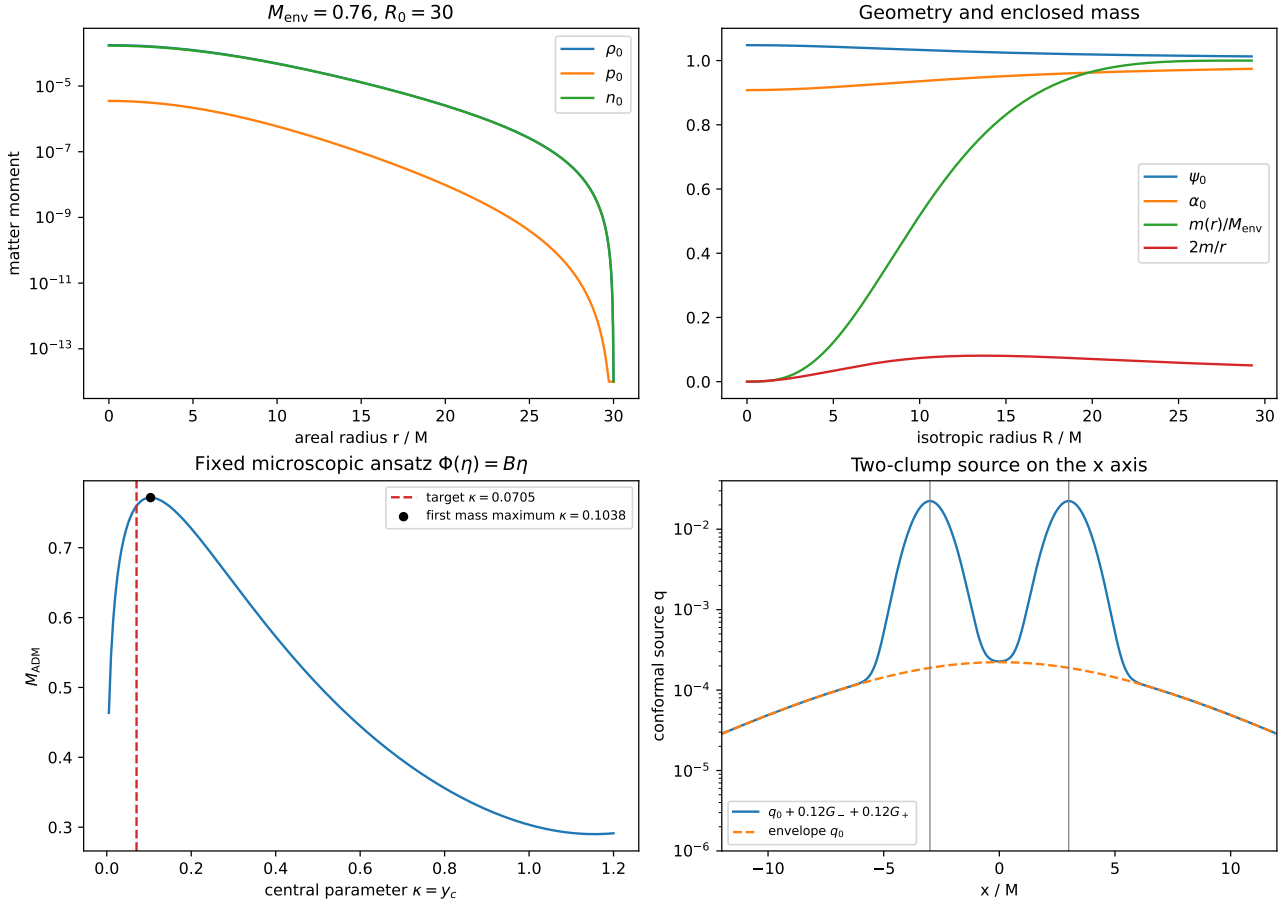


Figure 1: Independent equilibrium and two-clump construction. The first ADM-mass maximum on the fixed $\Phi(\eta) = B\eta$ sequence is at $y_c = 0.10375414$; the target is on its low-redshift side. This fact alone is not a stability theorem.

1.2 Stability statement and numerical gate

The linear stability result of Hadžić, Lin, and Rein is conditional on both a negative-Morse-index hypothesis for a reduced operator and a derivative-sign condition. We have not computed that operator for this particular member. Small-redshift results also do not provide a numerical threshold proving that $y_c = 0.0705$ lies in their regime. Therefore existence, weak compactness, and location before the first mass maximum are *not* promoted to an analytic stability certificate.

Instead, a cheap unperturbed realization used 500,000 particles, root $\Delta x = 1$, localized refinement through $\Delta x = 1/4$, and evolved to $t = 30$. All particles and rest mass were conserved. Energy-weighted r_{50} changed by $+3.64 \times 10^{-4}$, central lapse by -2.65×10^{-4}, and total $\sum mW$ by -7.18×10^{-5}. Enclosed energy changed by at most 0.58% at radii $2, 4, 8, \ldots, 30$; $\|H\|_2$ fell from 4.49×10^{-2} to 2.67×10^{-3} and $\|M\|_2$ from 6.74×10^{-3} to 8.68×10^{-4}. The central deposited cell is visibly PIC-noisy, but the $R < 4$ mean density changed only 1.88% and there is no secular radial contraction or expansion. This is the operational stability pass.

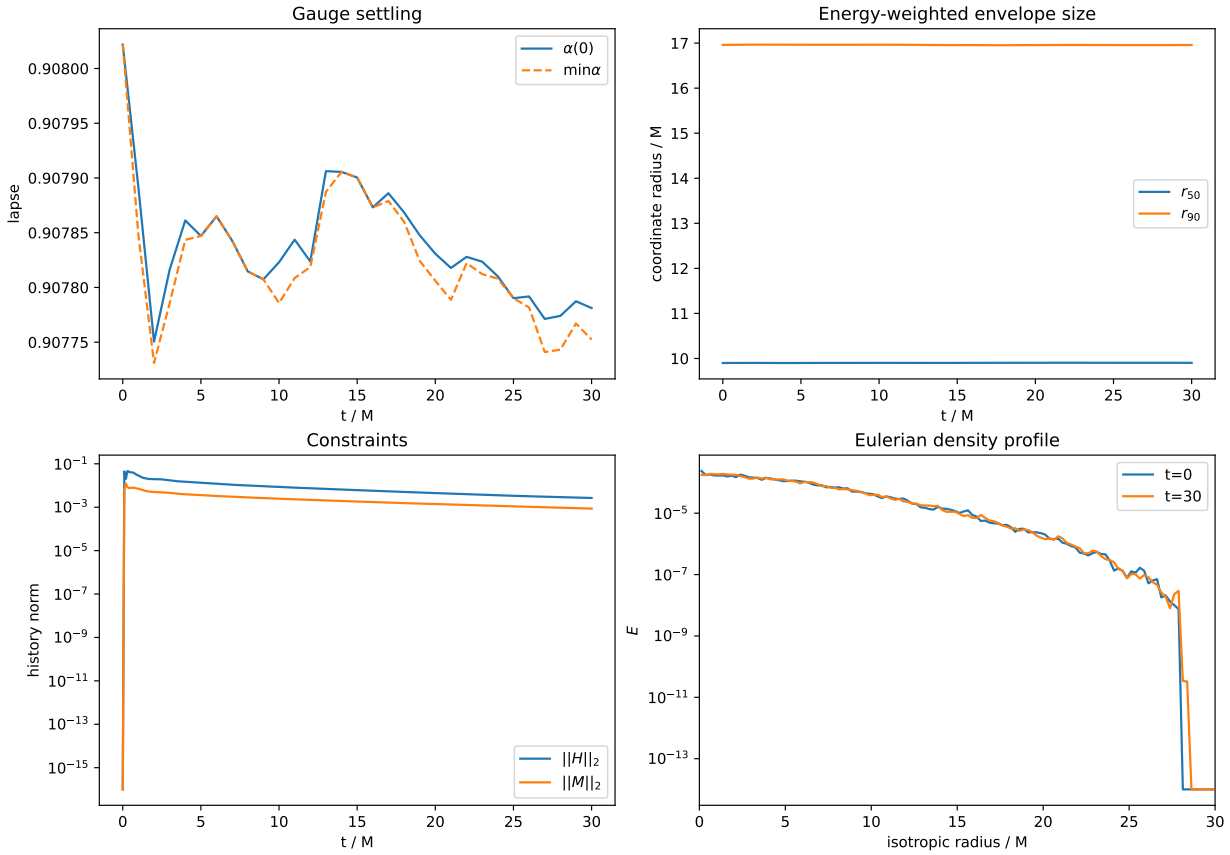


Figure 2: Unperturbed-envelope stability test. Grid-free particle radii and enclosed energy are authoritative for size evolution; the Eulerian density slice shows the expected single-cell PIC noise.

2 Two-clump conformal solve

Let $\boldsymbol{X}_\pm = (\pm 3, 0, 0)$, $M_+ = M_- = 0.12$, and $\sigma = 0.70$. With

$$G_a(\boldsymbol{X}) = \frac{\exp[-|\boldsymbol{X} - \boldsymbol{X}_a|^2/(2\sigma^2)]}{(2\pi\sigma^2)^{3/2}}, \quad q = q_0 + \sum_{a=\pm} M_a G_a, \quad \rho = \psi^{-5} q, \tag{7}$$

choose $K_{ij} = 0$ and initially $j^i = 0$. The Hamiltonian constraint is linear:

$$\nabla^2\psi = -2\pi q, \qquad \psi = \psi_0 + \sum_{a=\pm} \frac{M_a}{2r_a} \operatorname{erf}\left(\frac{r_a}{\sqrt{2}\sigma}\right), \quad r_a = |\boldsymbol{X} - \boldsymbol{X}_a|. \tag{8}$$

Since $\int q_0 \mathrm{d}^3X = 0.76$, $\int G_a \mathrm{d}^3X = 1$, and the mass dipole cancels,

$$\psi = 1 + \frac{0.76 + 0.12 + 0.12}{2R} + O(R^{-3}) = 1 + \frac{1}{2R} + O(R^{-3}), \tag{9}$$

which proves continuum $M_{\mathrm{ADM}} = 1$ without angular momentum. At either clump center $\psi = 1.124482509$, with envelope, clump, and total Eulerian densities 1.0518072×10^{-4}, 1.2355314×10^{-2}, and 1.2460495×10^{-2}, respectively. The initial lapse is $\alpha = \alpha_0(\psi_0/\psi)^2 = 0.7889962$ at a clump center.

3 Phase-space and five-million-particle realization

For the envelope,

$$f_{\mathrm{env}}(\boldsymbol{X}, \boldsymbol{v}) = \left(\frac{\psi_0}{\psi}\right)^5 f_0(R, \boldsymbol{v}), \tag{10}$$

which gives $\rho_{\mathrm{env}} = \psi^{-5} q_0$. Its position measure is $\psi\psi_0^5 n_0 \mathrm{d}^3X$; momenta are drawn from f_0/n_0. Thus

$$\mathcal{N}_{\mathrm{env}} = \int \psi\psi_0^5 n_0 \mathrm{d}^3X = 0.782171796044, \qquad \mu_{\mathrm{env}} = \mathcal{N}_{\mathrm{env}}/3{,}000{,}000 = 2.60723932015 \times 10^{-7}. \tag{11}$$

For each clump use an isotropic Maxwellian $h_a \propto \exp[-v^2/(2s^2)]$ with $s = 0.02$ and $\langle W\rangle = 1.00059970042$. Define

$$\begin{aligned} I_a &= \int \psi M_a G_a \mathrm{d}^3X = 0.132499635245, \\ f_a &= \frac{\psi^{-5} M_a G_a}{\langle W\rangle} h_a, \qquad \mu_a = \frac{I_a}{N_a\langle W\rangle} = 1.32420222782 \times 10^{-7}. \end{aligned} \tag{12}$$

with $N_a = 1{,}000{,}000$. Positions are sampled from ψG_a and every thermal draw is inserted as a same-position antithetic pair $(\boldsymbol{v}, -\boldsymbol{v})$. The two clumps use the established seeded sampler rather than an exact mirrored realization. In the 5M startup ledger their $\sum mW$ values differ by only 1.88×10^{-7} relative; the envelope and both clump currents cancel pairwise to floating-point roundoff.

4 Jeans, collapse, and horizon scales

Using the full new local density and metric at $R = 3$, the collisionless estimate is

$$k_J = \frac{\sqrt{4\pi\rho}}{s}, \qquad \lambda_{J,\mathrm{proper}} = \frac{2\pi}{k_J}, \qquad \lambda_{J,\mathrm{coord}} = \frac{\lambda_{J,\mathrm{proper}}}{\psi^2}, \tag{13}$$

and the stated Jeans-mass convention is a sphere whose diameter is λ_J:

$$M_J = \frac{4\pi}{3}\rho\left(\frac{\lambda_{J,\mathrm{proper}}}{2}\right)^3. \tag{14}$$

Numerically,

$$\lambda_{J,\mathrm{proper}} = 0.3175684, \quad \lambda_{J,\mathrm{coord}} = 0.2511492, \quad M_J = 2.0895163 \times 10^{-4}, \quad \frac{0.12}{M_J} = 574.3. \tag{15}$$

The clump diameter is $2\sigma = 5.57\lambda_{J,\mathrm{coord}}$, $t_{\mathrm{ff}} = \sqrt{3\pi/(32\rho)} = 4.862$, and the Newtonian Gaussian estimate gives $2T/|W| = 0.0248$: thermal support is negligible. At $\Delta x = 1/64$ the initial Jeans length spans 16.1 cells, so it is already well resolved.

Black-hole resolution is stricter. A $0.12M$ Schwarzschild hole has areal radius $2M = 0.24$ and isotropic horizon radius about $M/2 = 0.06$. This is 3.84, 7.68, and 15.36 cells at $\Delta x = 1/64, 1/128, 1/256$. The empirical FastFlow calibration found two cells unreliable, four marginal/intermittent, and eight or more substantially healthier. Hence 1/64 is sufficient for the initial Jeans problem but not the trusted horizon gate; 1/128 is the intended formation scale and 1/256 is the protected production maximum.

5 Actual Session-004 hierarchy and evolution

The box is $[-128, 128]^3$. Four root MeshBlocks per dimension, each with 32^3 cells and width 64, give exact root $\Delta x = 2$. Compact central safety regions and two towers use physical widths

$$2, 1, 1/2, 1/4, 1/8, 1/16, 1/32, 1/64, \tag{16}$$

without filling the domain at fine resolution. The priced initial mesh has 904 leaf blocks after 2:1 closure. Löhner refinement of $\alpha\psi^7$ at threshold 0.2 may create $\Delta x = 1/128$ and then the emergency maximum 1/256.

Evolution uses ordinary Z4c, RK4, CFL 0.4, standard advective $1 + \log$ slicing ("oplog" coefficient 2), the established shift with $\eta = 2$, dissipation 0.1, and Z4 damping $\kappa_1 = 0.02$, $\kappa_2 = 0$. No experimental Gamma-B driver, filtering, diffusion, or higher-order deposition experiment is enabled. Particle feedback uses conservative cross-level deposition and the GR-Boris pusher with fatal rejected-update protection plus the validated trilinear metric-interpolation fallback. Two individual FastFlow finders start at $t = 2.5$ and search every full step. AH publication requires finite positive on-mesh geometry, radius no smaller than four *local* cells, normalized expansion tolerance, and three persistent finds. Only published AHs excise particles; lapse excision is disabled.

The proposed Session-005 Vista box is $[-256, 256]^3$, with eight 32^3 root blocks per dimension so root Δx remains 2. It retains compact 1/64 initial towers, targets 1/128 adaptively, and permits 1/256 near each collapse. The final block, GPU, memory, walltime, and output counts are to be filled from the Session-004 evolution measurements rather than extrapolated from logical level numbers alone.

6 Orbital-angular-momentum construction

The intended signs for positive L_z are $\beta_y < 0$ at $x = -3$ and $\beta_y > 0$ at $x = +3$. In the spherical envelope at isotropic $R = 3$, the areal radius, enclosed mass, and pressure are $r = 3.28292793$, $m = 0.02379189$, and $p = 2.85580 \times 10^{-6}$. The local circular-geodesic estimate plus a Newtonian companion term is

$$v_{\mathrm{circ}}^2 \simeq \frac{m + 4\pi r^3 p}{r - 2m} + \frac{M_c(3)}{6^2} = 0.00774621 + 0.0100000, \qquad v_{\mathrm{circ}} = 0.133215. \tag{17}$$

One fiducial, not a scan, uses $\beta = 0.7 v_{\mathrm{circ}} = 0.09325043$.

Each antithetic thermal pair is formed in the clump rest frame and both members receive the same orthonormal Lorentz boost,

$$W' = \gamma(W + \beta v_{\hat{y}}), \qquad v'_{\hat{y}} = \gamma(v_{\hat{y}} + \beta W), \qquad v'_{\hat{x}} = v_{\hat{x}}, \quad v'_{\hat{z}} = v_{\hat{z}}, \tag{18}$$

with opposite β for the other clump. Stored covariant momenta are $u_i = \psi^2 v'_i$. Because $\langle W' \rangle = \gamma \langle W \rangle$, the energy-preserving particle weight is

$$\mu_a^{\mathrm{boost}} = \frac{I_a}{N_a \gamma \langle W \rangle} = 1.31843226262 \times 10^{-7}. \tag{19}$$

Thus the boost adds no avoidable Hamiltonian normalization error. The measured 5M ledger gives $P_{\hat{y}} = \mp 0.01235565$ per clump, zero summed orthonormal momentum, covariant $P_y^{\rm tot} = -2.6 \times 10^{-7}$, and $L_z^{\rm cov} = 0.0903221$.

6.1 Momentum-constraint defect and gate

The boosted clumps have local $j_{\hat{y}}/\rho_a = \pm\beta$ even though global linear momentum cancels. Therefore

$$D_j(K^{ij} - \gamma^{ij}K) = 8\pi j^i \tag{20}$$

is not solved by $K_{ij} = 0$. Matched 5M runs used identical positions, mesh, and seed. At $t = 0$ the local current has the expected opposite lobes; the first computed constraint slice at $t = 0.00625$ gives, in the $[-8, 8] \times [-4, 4]$ clump window,

	non-AM	boosted	ratio
H RMS	0.0966497	0.0966989	1.00051
$\lvert M\rvert$ RMS	0.00360761	0.00390603	1.08272
M_y RMS	0.0174054	0.0192321	1.10495

The maximum $|M|$ increases by 30.8%, localized at the clumps, while ordinary PIC/discretization residuals dominate the area-weighted RMS. The predeclared exploratory gate required less than 25% added local momentum RMS and less than 5% Hamiltonian RMS; it passes at 8.27% and 0.051%. This permits one clearly labeled exploratory evolution, but does not make the slice constraint-solved or suitable as a default production model.

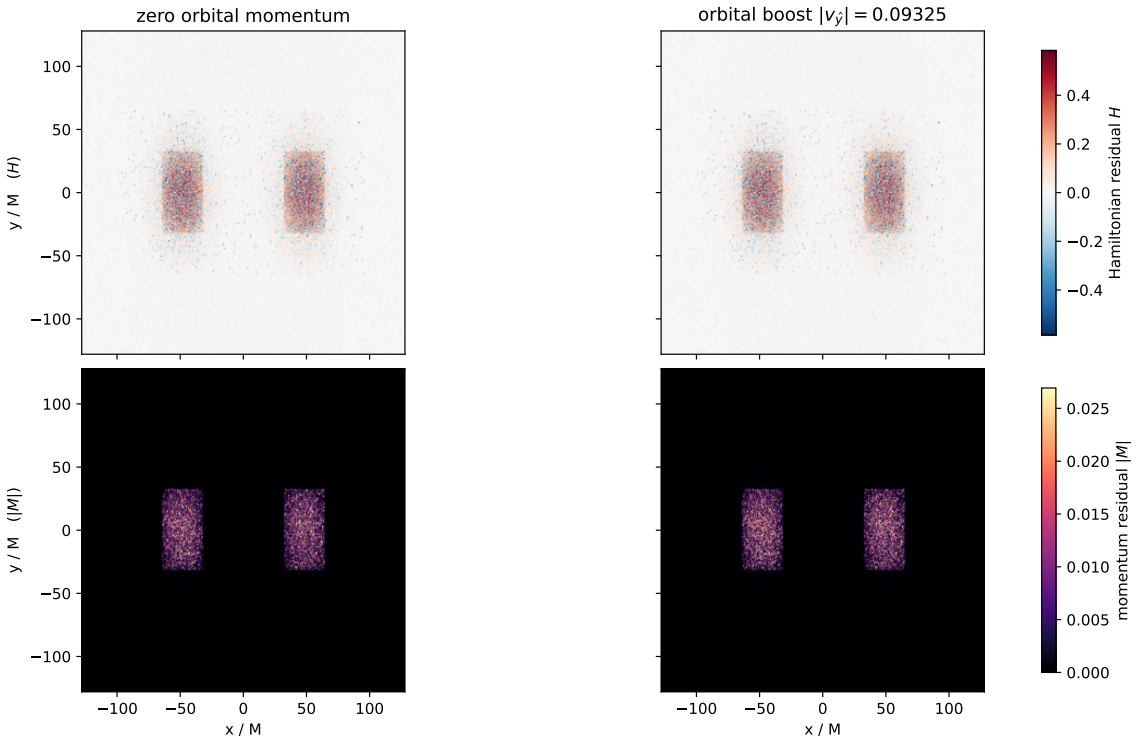


Figure 3: Direct matched-slice constraint comparison. The orbital current adds a localized momentum residual; the Hamiltonian residual is essentially unchanged.

References

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